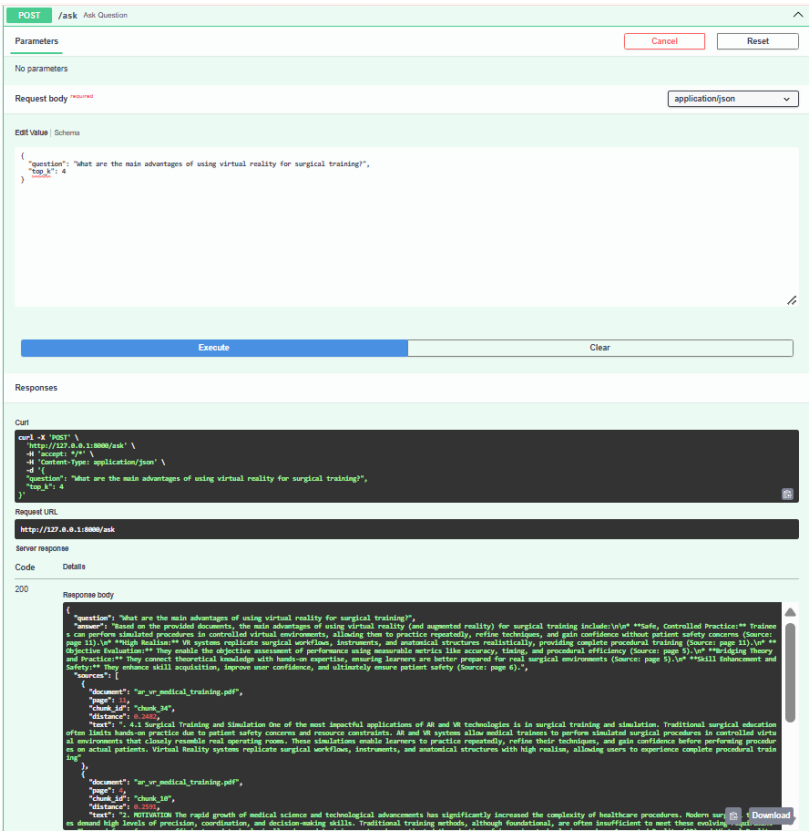
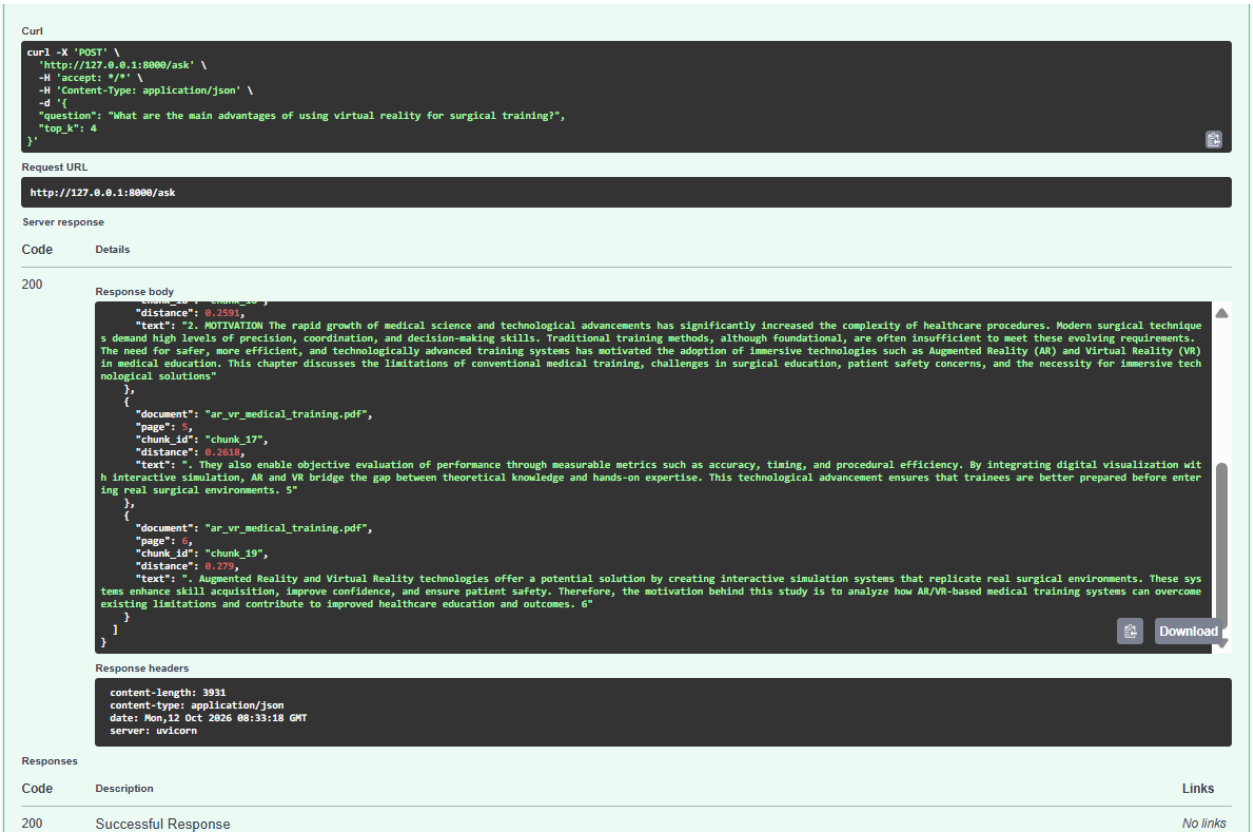
Final Mini Project – RAG Document Q&A: Test Screenshots

1. Test: Advantages of VR for surgical training

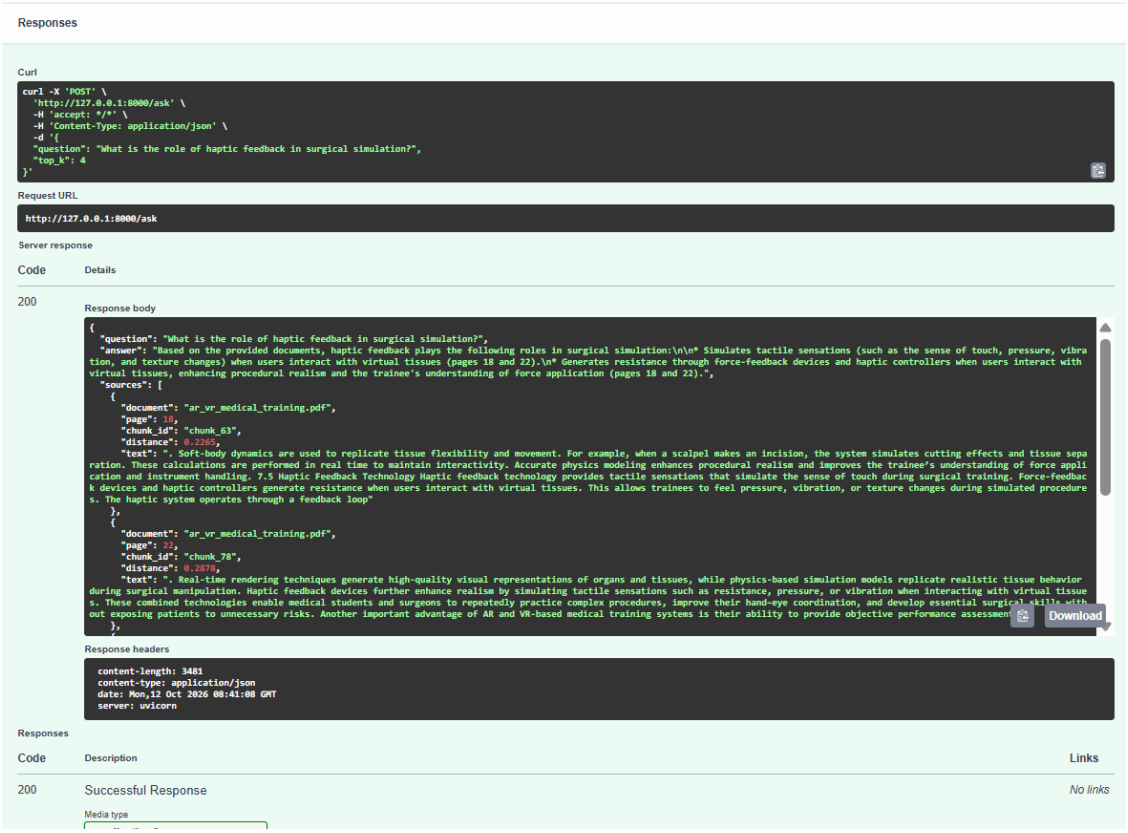
The answer lists safe practice, high realism, objective evaluation and skill enhancement with page citations, backed by close matches (distance ~0.25–0.28).

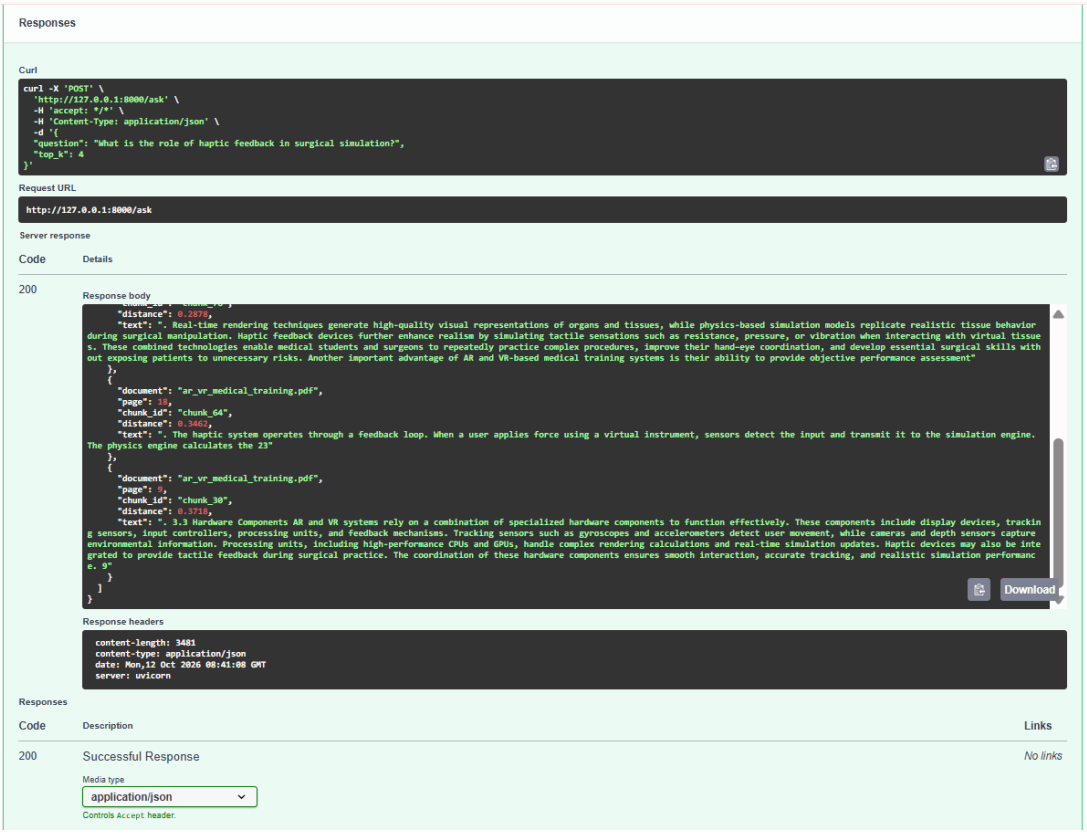




2. Test: Role of haptic feedback in surgical simulation

The answer explains tactile sensation and force feedback (pages 18 and 22), supported by closely matching chunks (distance ~0.23–0.37).





3. Test: Limitations of AR/VR in medical education

The answer identifies technical, financial and ethical challenges (page 20) from the retrieved limitations chunks (distance ~0.20–0.23).

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "question": "Limitations of AR/VR in medical education?",
  "top_k": 4
}'
```

Request URL

```
http://127.0.0.1:8000/ask
```

Server response

Code | Details

200

Response body


```
{
  "question": "Limitations of AR/VR in medical education?",
  "answer": "Based on the provided documents, AR and VR systems face technical limitations that affect performance and usability (page 20). Additionally, they face financial and ethical chal
lenges that must be addressed for widespread adoption (page 20).",
  "sources": [
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 20,
      "chunk_id": "chunk_69",
      "distance": 0.2011,
      "text": ". This enhances spatial understanding and makes complex medical concepts easier to comprehend. Another major advantage is repeatability, as trainees can practice procedures mu
ltiple times under identical conditions, which is not possible in real clinical environments. Additionally, AR/VR systems provide real-time feedback and performance analysis, allowing learne
rs to monitor progress and identify areas for improvement. Over time, these systems may also reduce training costs by minimizing reliance on physical resources such as cadavers and specializ
ed equipment. 8.2 Technical Limitations Despite their advantages, AR and VR systems face several technical limitations that affect performance and usability"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 23,
      "chunk_id": "chunk_81",
      "distance": 0.2155,
      "text": ". They also support medical education through interactive learning environments that simulate real-life medical situations. In rehabilitation therapy, VR-based systems help pa
tients recover from physical injuries or neurological disorders by providing engaging exercise environments that encourage consistent participation in treatment programs. Additionally, AR an
d VR technologies enable remote collaboration and telemedicine, allowing healthcare professionals from different geographical locations to interact, share knowledge, and assist in medical pr
ocedures.. In conclusion, AR and VR technologies represent a significant advancement in medical education and healthcare practice"
    },
    {
      "document": "ar_vr_medical_training.pdf",
```


Download

Response headers

```
content-length: 3382
content-type: application/json
date: Mon,12 Oct 2026 08:43:09 GMT
server: uvicorn
```

Responses

Code | Description | Links

200 | Successful Response | *No links*

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "question": "Limitations of AR/VR in medical education?",
  "top_k": 4
}'
```

Request URL

```
http://127.0.0.1:8000/ask
```

Server response

Code | Details

200

Response body


```
      "text": ". They also support medical education through interactive learning environments that simulate real-life medical situations. In rehabilitation therapy, VR-based systems help pa
tients recover from physical injuries or neurological disorders by providing engaging exercise environments that encourage consistent participation in treatment programs. Additionally, AR an
d VR technologies enable remote collaboration and telemedicine, allowing healthcare professionals from different geographical locations to interact, share knowledge, and assist in medical pr
ocedures.. In conclusion, AR and VR technologies represent a significant advancement in medical education and healthcare practice"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 7,
      "chunk_id": "chunk_20",
      "distance": 0.2225,
      "text": "3. FUNDAMENTALS OF AR AND VR TECHNOLOGY Augmented Reality (AR) and Virtual Reality (VR) are immersive technologies that integrate computer graphics, sensor systems, tracking a
lgorithms, and interactive software to enhance or simulate real-world environments. In medical training and surgical simulation, these technologies form the foundation for creating realistic
and interactive learning platforms. Understanding the fundamental concepts, operational principles, and technical components of AR and VR systems is essential for analyzing their effectivene
ss in healthcare education"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 20,
      "chunk_id": "chunk_67",
      "distance": 0.2337,
      "text": "8. ADVANTAGES AND LIMITATIONS Augmented Reality (AR) and Virtual Reality (VR) technologies have introduced significant improvements in medical training and healthcare practice
s by providing immersive and interactive learning environments. These technologies enhance visualization, improve skill acquisition, and enable safe simulation-based training. However, despi
te their numerous advantages, AR and VR systems also face technical, financial, and ethical challenges that must be addressed for widespread adoption. Understanding both the benefits and lim
itations of these technologies is essential for evaluating their effectiveness and feasibility in healthcare environments"
    }
  ]
}
```


Download

Response headers

```
content-length: 3382
content-type: application/json
date: Mon,12 Oct 2026 08:43:09 GMT
server: uvicorn
```

4. Test: Motion tracking in realistic surgical simulation

The answer describes sensor-based tracking and alignment with page citations, retrieved from the technical-working chunks (distance ~0.26–0.31).

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "question": "How motion tracking creates realistic surgical simulations?",
  "top_k": 4
}'
```

Request URL

```
http://127.0.0.1:8000/ask
```

Server response

Code | Details

200

Response body


```
{
  "question": "How motion tracking creates realistic surgical simulations?",
  "answer": "Based on the provided context, motion tracking systems create realistic surgical simulations by:\n\n* Capturing the movements of the user and translating them into corresponding actions within the virtual environment (pages 17 and 22).\n* Using sensors such as gyroscopes, accelerometers, infrared cameras, and depth sensors to detect head position, hand movement, and instrument orientation (page 17).\n* Processing tracking data to determine spatial coordinates and rotational angles (page 17).\n* Ensuring accurate interaction with virtual surgical tools and anatomical models (page 22).\n* Using advanced tracking algorithms to ensure precise alignment between physical movements and virtual responses, which prevents affecting the realism of instrument handling (page 18).",
  "sources": [
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 17,
      "chunk_id": "chunk_59",
      "distance": 0.2557,
      "text": ". Advanced systems incorporate deformable models that allow tissues to stretch, compress, or respond dynamically during simulated procedures. Accurate anatomical modeling ensures that the virtual environment closely represents real human anatomy, which is essential for effective surgical training. 7.2 Motion Tracking and Sensors Motion tracking systems are responsible for capturing user movements and translating them into corresponding actions within the virtual environment. AR/VR surgical simulators use sensors such as gyroscopes, accelerometers, infrared cameras, and depth sensors to detect head position, hand movement, and instrument orientation. The tracking data is processed to determine spatial coordinates and rotational angles"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 22,
      "chunk_id": "chunk_77",
      "distance": 0.278,
      "text": ". VR systems, on the other hand, create fully immersive digital environments that replicate real operating rooms and medical scenarios, allowing trainees to perform simulated procedures in a controlled and risk-free setting. Advanced technologies such as three-dimensional anatomical modeling, motion tracking, real-time rendering, physics-based simulation, and haptic feedback play an important role in making these simulations realistic and effective. Motion tracking systems capture the movements of the user and translate them into corresponding actions in the virtual environment, ensuring accurate interaction with virtual surgical tools and anatomical models"
```


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Response headers

```
content-length: 4148
content-type: application/json
date: Mon,12 Oct 2026 08:49:24 GMT
server: uvicorn
```

Responses

Code | Description | Links

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/ask' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
  "question": "How motion tracking creates realistic surgical simulations?",
  "top_k": 4
}'
```

Request URL

```
http://127.0.0.1:8000/ask
```

Server response

Code | Details

200

Response body


```
      "text": ". VR systems, on the other hand, create fully immersive digital environments that replicate real operating rooms and medical scenarios, allowing trainees to perform simulated procedures in a controlled and risk-free setting. Advanced technologies such as three-dimensional anatomical modeling, motion tracking, real-time rendering, physics-based simulation, and haptic feedback play an important role in making these simulations realistic and effective. Motion tracking systems capture the movements of the user and translate them into corresponding actions in the virtual environment, ensuring accurate interaction with virtual surgical tools and anatomical models"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 17,
      "chunk_id": "chunk_57",
      "distance": 0.2825,
      "text": "7. TECHNICAL WORKING OF SURGICAL SIMULATION The technical working of AR/VR-based surgical simulation systems involves the integration of advanced computational models, real-time tracking mechanisms, rendering engines, and feedback systems. These technologies collectively create immersive environments that replicate real surgical procedures with high precision. The system processes medical data, interprets user input, simulates anatomical behavior, and generates interactive visual and tactile responses. This chapter explains the core technical components that enable realistic surgical simulation. 7.1 3D Anatomical Modeling Three-dimensional anatomical modeling forms the foundation of surgical simulation systems"
    },
    {
      "document": "ar_vr_medical_training.pdf",
      "page": 18,
      "chunk_id": "chunk_61",
      "distance": 0.3053,
      "text": "tracking may affect the realism of instrument handling. Therefore, advanced tracking algorithms are used to ensure precise alignment between physical movements and virtual responses. 7.3 Real-Time Rendering Techniques Real-time rendering is the process of generating visual output instantly based on user interaction. Surgical simulation systems require high frame rates to ensure smooth visualization and prevent motion sickness. The rendering pipeline processes 3D models, lighting effects, textures, and shading calculations to produce realistic images displayed through AR glasses or VR head-mounted displays. Graphics Processing Units (GPUs) perform complex calculations required for rendering anatomical structures and surgical tools"
    }
  ]
}
```


Download

Response headers

```
content-length: 4148
content-type: application/json
date: Mon,12 Oct 2026 08:49:24 GMT
server: uvicorn
```

Responses

Code | Description | Links